

Research Internship Application & Technical Portfolio Dossier

Computational Structural Dynamics, Physics-Informed Neural Networks, and Real-Time Earthquake Engineering

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Repository: https://github.com/raghvendra1gdsc-png/seismic_ai • **Verification:** 77 Passing Automated Tests

1. Personal Statement & Research Motivation

Dear Professor,

I am writing to express my earnest interest in joining your research group as a student research intern. Over the past several months, I have been dedicated to designing and building **Seismic-AI**, an open-source computational framework that explores how classical nonlinear structural mechanics and physics-informed machine learning can be brought together to solve a high-stakes engineering problem: **the earthquake lead-time gap**.

Why I Chose This Problem: During my study of structural dynamics, I observed a frustrating trade-off in current practice. On one hand, high-fidelity numerical time-history analysis (such as implicit Newmark- β integration with Newton-Raphson nonlinear equilibrium) is mechanically rigorous and trustworthy, but it is far too slow for real-time cyber-physical early warning. Even for a modest multi-storey frame, numerical step-by-step convergence requires seconds to minutes—far exceeding the critical 10 to 30 second window between the arrival of non-destructive primary compressional waves (P -waves) and destructive transverse shear waves (S -waves). On the other hand, standard 'black-box' deep learning models can execute in microseconds, but they have no intrinsic understanding of physics. When confronted with rare, high-amplitude near-fault ground motions outside their training data, purely data-driven models frequently predict impossible floor drifts, violate dynamic equilibrium ($\Sigma F \neq ma$), and exhibit unphysical energy growth.

My Research Rationale: I believe that resilient civil infrastructure requires *Physics-Informed Scientific Machine Learning (SciML)*. Rather than relying on blind regression, we can enforce Newton's second law, Bouc-Wen hysteretic constitutive relationships, and energy conservation directly inside the neural network's loss function. This allows the neural surrogate to retain the mechanical rigor and conservation guarantees of continuum mechanics while achieving over **60,000× computational acceleration** (**< 0.5 μ s inference**). This microsecond speed makes real-time cyber-physical early warning and building-level safety actuation physically viable.

How I Wish to Contribute to Your Laboratory: In your research group, I am eager to contribute my strong foundations in solid mechanics, numerical mathematics, and software craftsmanship. Specifically, I wish to collaborate with you on: (1) *3D Continuum Neural Operators*: extending lumped-mass models to full 3D solid mechanics and soil-structure interaction using Fourier Neural Operators (FNO) and DeepONets; (2) *Online Bayesian System Identification*: combining streaming sensor telemetry with unscented Kalman filters to track progressive hysteretic stiffness degradation in real time during aftershock swarms; (3) *Embedded Edge Hardware Deployment*: quantizing neural surrogates onto low-power ARM/FPGA microcontrollers directly integrated with on-structure MEMS sensors; and (4) *Academic Publication*: assisting in drafting and submitting rigorous, peer-reviewed research papers to premier structural engineering journals such as *Earthquake Engineering & Structural Dynamics* (EESD).

2. What I Have Built: Core Engineering & Scientific Contributions

1. Inelastic Structural Dynamics Engine: Formulated multi-degree-of-freedom (MDOF) dynamic equilibrium from first principles, assembling lumped floor mass matrices, tridiagonal shear stiffness matrices, and classical Rayleigh damping. Built a vectorized Newmark- β implicit solver ($\gamma=0.5$, $\beta=0.25$) coupled with a 13-parameter Bouc-Wen nonlinear hysteretic differential model resolved via Newton-Raphson equilibrium iterations to a residual tolerance of $< 10^{-6}$ kN.

2. Physics-Informed Neural Network (PINN) Surrogate: Designed and trained a multi-layer neural surrogate with a composite loss function: $L_{\text{total}} = L_{\text{data}} + \lambda_{\text{eq}} L_{\text{equilibrium}} + \lambda_{\text{energy}} L_{\text{energy}}$. The model predicts peak interstorey drift ratios (PIDR) and base shear in $< 0.5 \mu\text{s}$ (**$> 60,000\times$ faster than step-by-step numerical ODE solvers**), maintaining $R^2 = 0.9510$ under strict dual-blind generalization testing across unseen earthquakes and unseen buildings.

3. Sensor Hardware Abstraction Layer (HAL) & Real-Time STA/LTA Trigger: Built a production-grade driver interface that ingests live triaxial acceleration at 100 Hz from physical USB/Serial MEMS sensors (ADXL355, MPU6050) and network MQTT IoT seismographs. Implemented an online 4th-order Butterworth bandpass filter (0.1–25 Hz) and a recursive Allen STA/LTA event detector with an average *P*-wave detection latency of **< 50 ms**.

4. Park-Ang Damage Evaluation & Emergency Actuation: Integrated the cumulative Park-Ang damage index (combining peak displacement and hysteretic energy dissipation). Automated safety threshold logic (Safe, Moderate, Evacuate, Collapse) triggers instant local network emergency sirens and industrial relay commands (e.g. parking elevators, closing main gas valves) in < 100 ms—well before destructive *S*-waves arrive.

5. Multi-Objective Resilience Optimization & Compliance Engine: Integrated NSGA-II Pareto optimization to discover the optimal trade-off between structural retrofit cost and FEMA P-58 collapse probability under Incremental Dynamic Analysis (IDA). Included automated 3-column engineering audits benchmarking prescriptive codes (IS 1893:2016 / ASCE 7-22) against nonlinear solvers and AI surrogates.

3. Theoretical Formulations & Governing Equations

A. MDOF Equations of Motion: Dynamic equilibrium for an *N*-storey shear building under ground acceleration $a_g(t)$ is governed by:

$$\mathbf{M} \ddot{\mathbf{u}}(t) + \mathbf{C} \dot{\mathbf{u}}(t) + \mathbf{F}_s(\mathbf{u}, \mathbf{z}, t) = -\mathbf{M} \mathbf{r} a_g(t) \quad (\text{Eq. 1})$$

where $\mathbf{M}$ is the lumped floor mass matrix, $\mathbf{C} = \alpha_R \mathbf{M} + \beta_R \mathbf{K}_0$ is the Rayleigh damping matrix, and $\mathbf{F}_s$ is the restoring force vector.

B. 13-Parameter Bouc-Wen Inelastic Hysteresis: Restoring force at storey *i* decomposes into elastic and hysteretic components:

$$f_{s,i} = \alpha_i k_i d_i + (1 - \alpha_i) k_i z_i \quad (\text{Eq. 2})$$

where $d_i = u_i - u_{i-1}$ is the drift, and z_i evolves according to the nonlinear differential equation:

$$\dot{z}_i = A_i d_i - \beta_i |d_i| |z_i|^{n-1} z_i - \gamma_i d_i |z_i|^n \quad (\text{Eq. 3})$$

Strength and stiffness degradation parameters scale dynamically with cumulative hysteretic energy: $E_H = \int (1 - \alpha) k z d\mathbf{u}$.

C. Park-Ang Damage Formulation: Structural damage is quantified using the classical Park-Ang index:

$$DI = (\mathbf{u}_m / \mathbf{u}_u) + (\beta_{PA} / (Q_y \mathbf{u}_u)) \int dE_h \quad (\text{Eq. 4})$$

where u_m is maximum displacement, u_u is monotonic capacity, Q_y is yield force, and $\beta_{PA} \approx 0.05$. When $DI \geq 0.40$, the structure transitions into severe inelastic damage, triggering mandatory building evacuation.

4. Scientific Benchmarks & Validation Results

To rigorously assess model generalization and eliminate data leakage, models were evaluated across a **4-Tier Generalization Protocol**:

Protocol Tier	Linear Ridge (R^2)	Random Forest (R^2)	Gradient Boosting (R^2)	PINN Neural Surrogate (R^2)
Tier 1: Random Split	0.9848	0.9340	0.9589	0.9850
Tier 2: Unseen Earthquakes	0.9724	0.8775	0.9463	0.9256
Tier 3: Unseen Buildings	0.9664	0.8865	0.9841	0.9426
Tier 4: Dual-Blind Unseen	0.8748	0.7244	0.9425	0.9510

Validation Against Published Literature (SAC Steel Project FEMA-355C):

- **SAC 3-Story LA Frame**: Experimental period $T_1 = 1.010$ s; Seismic-AI Modal Engine $T_1 = 1.012$ s (**0.20% error**).
- **SAC 9-Story LA Frame**: Experimental period $T_1 = 2.270$ s; Seismic-AI Modal Engine $T_1 = 2.268$ s (**0.09% error**).
- **Computational Latency**: Full step-by-step solver takes 32.4 ms; PINN surrogate completes in **0.00048 ms** (> 60,000× acceleration).

5. End-to-End Walkthrough: Timeline from P-Wave to Evacuation

1. **Continuous Sensing ($t < 0$ s)**: Serial MEMS driver streams 100 Hz triaxial acceleration. Butterworth bandpass filter removes high-frequency noise. STA/LTA ratio remains baseline ($r \approx 1.0$ – 1.2).
2. **P-Wave Detection ($t = 0.038$ s)**: Primary wave arrives. Energy ratio r spikes to ≥ 3.5 . Trigger latches event timestamp and peak spectral ordinates in < 50 ms.
3. **Surrogate Prediction ($t = 0.039$ s)**: Features pass to PINN surrogate. Forward inference finishes in 0.00048 ms, predicting peak drift (PIDR = 1.42%) and base shear.
4. **Damage Assessment ($t = 0.040$ s)**: Park-Ang index evaluated (DI = 0.62). Threshold check indicates Severe Inelastic Damage, initiating EVACUATION state.
5. **LAN Siren & Actuation ($t = 0.055$ s)**: Webhooks and WebSockets broadcast emergency commands. Browser sirens chime and relays park elevators and cut gas.
6. **S-Wave Arrival ($t = 14.200$ s)**: Destructive shear waves strike. Occupants have had 14.14 seconds of advance warning; facilities are protected.

6. Proposed 3-Month Research Roadmap

Month / Milestone	Research Objective	Target Deliverable
Month 1: 3D Continuum FNO	Extend MDOF surrogates to 3D continuum finite element models and soil-structure interaction.	Trained Fourier Neural Operator resolving 3D stress concentrations.
Month 2: Online System ID	Couple streaming telemetry with unscented Kalman filters for real-time damage tracking.	Online stiffness identification algorithm tested on multi-event sequences.
Month 3: Edge & Publication	Quantize surrogates for low-power ARM/FPGA microcontrollers; draft journal paper.	Sub-millisecond on-sensor benchmark and co-authored ASCE/EESD paper draft.

Thank you very much for your time and consideration of my application. I would welcome the opportunity to discuss my work, answer any technical questions, or walk through the implementation in an interview.

Sincerely,

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